Station 1 (24 pts)

Identify specimens A-C. [3]

For each specimen, indicate intrusive (I) or extrusive (E). [3]

For each specimen, indicate composition: felsic (F), intermediate (I), or mafic (M). [3]

Rank A-C from highest to lowest silica content. [3]

Label A-C as part of a Bowen's Reaction Series temperature range. [3]

1. Which specimen is most likely to form near a subduction zone? Why? [2]
2. Rank specimens A-C from highest to lowest eruption explosivity. [3]
3. Which specimen would crystallize olivine first? Why? [2]
4. Explain why specimen A typically forms darker-colored rocks than specimen C. [2]

Station 2 (18 pts)

Identify specimen A and B. [2]

Classify each as clastic, biochemical, or chemical. [2]

Identity each specimen's depositional environment [2]

1. Which indicates higher energy deposition? Explain. [3]
2. Describe grain rounding and sorting for each specimen. [4]
3. Which specimen is more compositionally mature? Why? [3]
4. Name one sedimentary structure likely associated with each rock. [2]

Station 3 (21 pts)

C:



D:



Identify specimens A-D. [4]

Indicate whether each is foliated or non-foliated. [4]

Identify the protolith of each specimen. [4]

Identify the specimen as low, medium, or high metamorphic grade. [4]

1. Which specimen shows compositional banding? Why does it form? [3]
2. Which specimen would most strongly effervesce in HCl (powdered)? [2]

Station 4 (16 pts)

1. Identify the specimen. [2]
2. Describe its biochemical origin. [2]
3. Rank bituminous, peat, lignite, anthracite by increasing coal rank. [4]
4. Explain how temperature and pressure change coal rank. [2]
5. State two economic uses of this specimen. [2]
6. Identify one environmental concern related to its extraction or use. [2]
7. What is this specimen’s depositional environment? [2]

Station 5 (15 pts)

1. Identify the mineral. [1]
2. State its mineral family. [1]
3. Give its Mohs hardness. [1]
4. State the chemical formula [2]
5. Describe the environment in which it forms. [2]
6. Explain why evaporite minerals indicate arid climates. [2]
7. Name the other evaporite mineral from the official list. [2]
8. What country has the world’s largest known exploitable deposits of this mineral? [2]
9. What are two major uses of this mineral? [2]

Station 6 (15 pts)

Identify specimens A-C. [3]

Classify each as metallic or nonmetallic. [3]

Give the streak color of each. [3]

1. Which specimen is commonly mistaken for gold? Why? [2]
2. List two minerals that are commonly found with specimen C. [2]
3. Which specimen conducts electricity? Why? [2]

Station 7 (15 pts)

Identify both specimens. [2]

Give the chemical formulas of both specimen [2]

Identity the Mohs hardness of both specimen [2]

1. Describe the acid reaction for specimen A. [2]
2. Identify the optical property observed in specimen A. [2]
3. Explain how biological activity contributes to the formation of specimen B. [3]
4. Which specimen would metamorphose into marble? [2]

Station 8 (16 pts)

Identify both minerals. [2]

State the mineral group for each. [2]

Identify both specimen's luster. [2]

Identify both specimen's specific gravity. [2]

1. Which mineral commonly forms dodecahedral crystals? [2]
2. Which mineral indicates copper-rich environments? [1]
3. Which mineral may be used as a gemstone? [1]
4. Explain the formation of specimen A [4]

Station 9 (16 pts)

Identify both specimens by name and as a rock/mineral. [4]

Give the chemical formula of specimen A and major component of B. [2]

1. Identify the mineral family of specimen A. [2]

2. Explain how feldspar appears/looks within specimen B. [2]

3. List the three major elements in the solid solution relationship among feldspars. [3]
   (**Div C only)**

4. Which feldspar endmember is sodium-rich? [1]
   (**Div C only)**

5. List two uses of specimen B. [2]

Station 10 (20 pts)

Identify specimens A–E. [5]

Identify each mineral's silicate structure type. [5] **(Div C Only)**

1. Which mineral has the lowest hardness? Why? [2]
2. Which mineral crystallizes at the highest temperature? Why? [2]
3. Which specimen is the most common mica? [2]
4. What is the optical property of specimen C? [2]
5. What is the crystal habit of specimen B? [2]

Station 11 (15 pts)

Identify both specimen. [2]

Identify the luster of both specimens. [2]

Identify the specific gravity of both specimen. [2]

1. Describe the role of dissolved gases in forming vesicles. [3]
2. Which specimen formed from the most viscous magma? Explain. [2]
3. Which specimen is felsic and highly vesicular? [2]
4. Which specimen is associated with explosive eruptions? Why? [2]

Station 12 (20 pts)

A



B



C

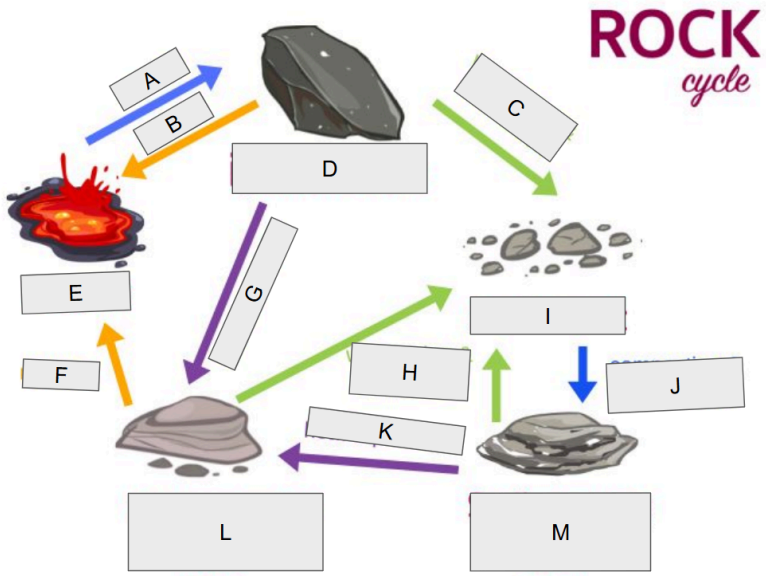


D



1. Identify the four bed forms [4]
2. Identify each structure's depositional energy regime. [4]
3. Which structure indicates periodic exposure to air? Explain. [3]
4. Which structure best indicates paleocurrent direction? Why? [3]
5. Explain how graded bedding helps determine stratigraphic "top" and "bottom". [4]
6. Rank the four based on preservation potential (lowest to highest) [2]

Station 13 (24 pts)



1. Label all major processes in the rock cycle. [7]

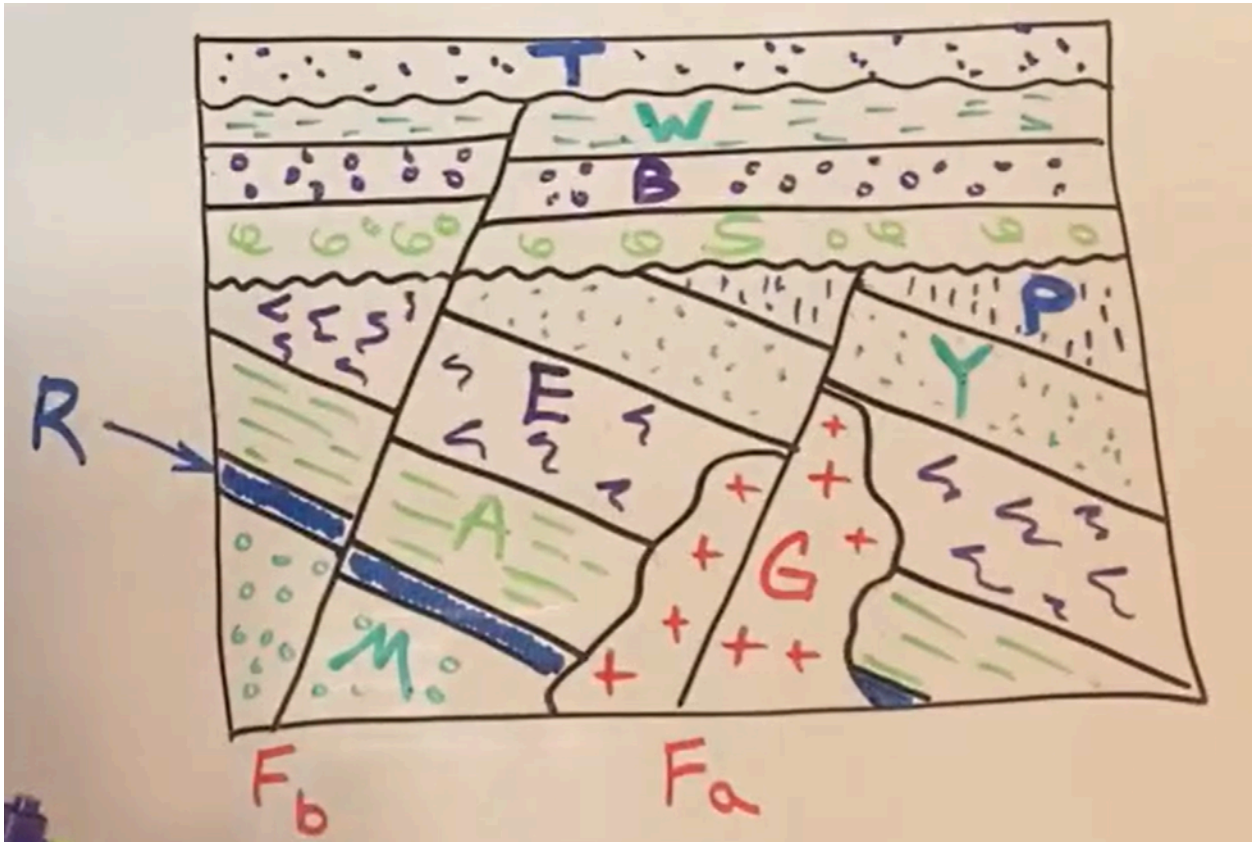
2. Identify one part of the rock cycle that is most strongly influenced by climate and explain why. [3]

3. Explain why the rock cycle is considered a dynamic system rather than a closed loop. [4]

4. Predict how the rock cycle would differ on a planet without plate tectonics. [4]

5. A shale undergoes burial, heating, and partial melting.
   a. Identify the rock types it may become at each stage. [3]
   b. Identify the dominant process at each stage. [3]

Station 14 (25 pts)



1. List the letters in the diagram from youngest to oldest [13]

2. What break is below S? [2]

3. Bed B and W display the principle of _____ [2]

4. What principle tells the relative age of G vs E? [2]

5. If fossils were found in two different layers, explain how **faunal succession** could be used to refine relative ages. [3]

6. Explain one limitation of relative dating and how absolute dating could address it. [3]

Station 15 (25 pts)

For this station, identify the rock described, then answer the related question.

1. This igneous rock forms from magma with extremely high silica content and viscosity. It commonly contains visible quartz and feldspar phenocrysts in a fine-grained groundmass and is chemically equivalent to granite. [3 pts]

2. In what tectonic setting is this rock most likely to form, and why is its eruption style typically explosive? [2 pts]

3. This metamorphic rock forms from shale under low-grade regional metamorphism. It exhibits a silky sheen caused by microscopic mica crystals and represents a transitional stage between slate and schist. [3 pts]

4. What metamorphic process causes the alignment of minerals that produces this rock's foliation? [2 pts]

5. This sedimentary rock is composed almost entirely of calcite cemented shell fragments. It commonly forms in warm, shallow marine environments and can later metamorphose into marble. [3 pts]

6. What biological process is primarily responsible for the formation of this rock? [2 pts]

7. This ultramafic rock originates in Earth's mantle and is composed primarily of olivine and pyroxene. When exposed at Earth's surface, it frequently alters to serpentinite through hydration reactions. [3 pts]

8. What plate tectonic process most commonly brings this rock to Earth's surface? [2 pts]

9. This clastic sedimentary rock is characterized by alternating layers of sandstone and shale deposited by turbidity currents. It is most commonly associated with deep-marine environments near continental margins. [3 pts]

10. Why is this rock particularly useful for reconstructing ancient submarine slope environments? [2 pts]

Station 16 (20 pts) - TB

1. Identify minerals 1-12. [12]
2. Identify rocks 13-20. [8]






















BullSO, University of South Florida - Repaired practice edition

Division C | Rocks And Minerals | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

repair-case. (2 points) Authored written substitute for unavailable source material: The best identification uses…

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

A. Only specimen size

B. Several diagnostic observations

C. Color alone

D. One nickname

Answer: _____